Silent-Failure Study

Status as of 31 August 2026 · **measurement complete**

Where you are: three problem families, three engines, an independent referee that shares no library with any of them.

(A) Engines can report success while returning wrong physics — SUPPORTED. (B) Engines *differ* in whether they do — SUPPORTED, once, narrowly (Drake's constraint API; see below). Everywhere else all three engines behave identically and correctly.

The three families

Family	New machinery	Degeneracy	Result
1. Pendulum chain	baseline; 1–30 links	amplitude drives it chaotic ($10^{11}\times$ growth)	all agree
2. Slider–crank	prismatic joint, closed loop	dead centre: $\mathrm{d}x/\mathrm{d}\theta \to 0$, inertia collapses $10^6\times$	all agree
3. Cart–pole	mixed joints in a tree, coupled $M(q)$	light cart: $\det M$ collapses $10^6\times$ at vertical	all agree

72 engine–case pairs across families 2 and 3: 0 disagreements, 0 refusals, worst residual 1.3×10^{-14}, on derivation *and* integration, including at the degenerate configurations.

The one engine-specific defect

! **Drake 1.56.0 accepts a distance constraint on a *continuous* plant and silently omits it.** Mass matrix comes back diagonal; slider acceleration exactly 0.0 against −0.993. No exception, no warning. Discrete mode is correct.
Provenance: guard present v1.11.0–v1.50.0 (#18196), removed by #24079 on 2026-02-17, absent v1.51.0 onward. Removal was *deliberate* — to enable CENIC — and the maintainers flagged the risk themselves, adding a throw to `EvalTimeDerivatives` but not yet to `CalcMassMatrix`, `CalcBiasTerm`, `CalcGravityGeneralizedForces`. An incomplete migration, not carelessness.
Worth: an upstream issue. Not a paper on its own.

Capability, where they genuinely differ

Largest chain answered	SymPy (idiomatic)	MechanicsDSL	Drake
Lagrangian	5	12	**30** (flat 0.23 s)
Hamiltonian	—	3	—

Every engine that answered, answered correctly; every engine that could not, timed out. **They differ in reach, not in honesty.**

The transferable result

! **An error that is exactly 1.00 or exactly 2.00 across every case is a convention mismatch in *your comparison*, not a defect in the engine.** Three independent confirmations: canonical (q, p) state vs. accelerations; Drake's relative joint angles vs. absolute; a mirrored pole axis. Second tell: the error vanishes in a degenerate case (N=1, M=I, θ=0) and grows away from it. Skipping this check would have produced three fabricated findings against widely used software.

Rules still in force

- ! Engine, scoring path, and the 55-case matrix frozen at `a8dc2b2`. Harness may grow; gate is the frozen matrix still returning 44/0/1/10 bit-identically.
- ! **Do not claim MechanicsDSL outperforms Drake.** On the cart–pole Drake's residuals are the *smallest* of the three (10^{-16}). Reaching 12 links against idiomatic SymPy's 5 is defensible; Drake reaches 30.
- ! Scope every Drake claim to **1.56.0**: master was read, not run, and #24079 implies behaviour there is integrator-dependent.

What is left

- ☐ **File the Drake issue** using `finding_drake_constraint.py` (standalone; `numpy` + `pydrake` only). Ask whether the three accessors are on #24079's manifest — do not assert a defect. Flat, descriptive, no adjectives; file before publishing.
- ☐ **Write it up.** Null result with strong methodology, one narrow engine-specific defect, a three-tier capability comparison, and the diagnostic above. Short or workshop paper, as `SCOPE.md` always said.

Reports: TR-2026-01 referee · 02 SymPy · 03 Drake · 04 Week 1 findings · 05 three families. **Disclose:** you maintain one of the engines measured; AI assistance was used. Both permitted, concealing either is not.